

AKSHAT SHRINATE

[LinkedIn](#) | akshatshrinате2002@gmail.com | +91-9555608551 | [GitHub](#) | Chandigarh, India

Education

Little Flower School (Gorakhpur, Uttar Pradesh) Class X, ICSE	2018 – 2019
• Among the top 2% students of a batch of 400+ students	Score: 94%
Academic Global School (Gorakhpur, Uttar Pradesh) Class XII, CBSE	2020 – 2021 Score: 85%
Punjab Engineering College - PEC (Chandigarh) B.Tech in Electronics and Communication Engineering (minor in Digital VLSI Technologies)	2021 – 2025 CGPA: 7.35

Projects

- Wi-Fi enabled ESP 32-based Detachable PowerDrive for manual wheelchairs** ([Link](#)) May 2025
- Developed an assistive detachable drive system for manual wheelchairs, which mounts onto the rear wheels to provide motorised mobility support for users, accommodating loads up to **50kg**.
 - Integrated a Wi-Fi-enabled, mobile-based application with voice control into the ESP32-based system, facilitating seamless user interaction and achieving a travel range of up to **6.75 km** per charge from a **24V-10Ah** battery.
- PIKA: Bio-Inspired Quadruped Locomotion** ([Link](#)) June 2025
- Implemented a Python-based locomotion framework from a biomechanics study by translating analytical gait equations into synchronised hip, knee, and paw joint trajectories for **5+** canine gait cycles in the sagittal plane.
 - Validated the generated trajectories through simulation and deployed them on the PIKA quadruped robot, enabling stable execution of multiple biologically inspired gait patterns.

Internships and Work Experience

- Copperpod IP – Research Analyst Intern** Jan 2024 – June 2024
- Conducted a detailed breakdown and analysis of **60+** patents across domains such as Virtual Reality Headsets, Artificial Intelligence, and Stereoscopic Computer Vision from **15+** clients to identify potential infringement cases.
 - Developed Backend architecture and APIs for a Web-based Communication Portal, enabling seamless uploading and downloading of deliverables across projects to enhance client collaboration and streamline project management.
- Next-Gen BIRD Lab, Indian Institute of Technology (IIT) Jodhpur** Sept 2025 – June 2026
- Developed **Robo-View**, an open-source web-based simulator that allows users to parse URDF models, execute forward kinematics, and visualize a robot's configuration space directly in the browser, eliminating the need for local ROS 2 or native RViz installations.
 - Designed and developed the embedded control system for a Twisted String Actuator (TSA) based dexterous robotic gripper, integrating computer vision to enable hand-movement mimicking and implementing voice control to switch between diverse grasping positions dynamically.

Technical Knowledge and Soft Skills

Technical Skills: Python, NodeJS, MongoDB, ROS 2, C++, Fusion 360, MySQL, NumPy, Pandas, Matplotlib, Scikit-Learn, Flask, Streamlit, Microsoft Excel, PowerPoint, Word, GitHub, Object-Oriented Programming (OOP), Database Management System (DBMS)

Patents

- Title: A Robotic Hand Assembly**
Indian Patent Application No. 202611086086, filed July 2026 (Pending).

Work Experience and Extra-Curricular/Leadership

- PECFEST Core Committee Head** Sept 2022 – Nov 2024
- Directed back-end operations for North India's largest techno-cultural festival, collaborating over **60+** days to ensure seamless execution.
 - Organized **10+** high-profile technical competitions and generated **165k+** impressions on Instagram, achieving widespread participant participation.

Achievements

- Campus Winner – Bajaj Ohm Challenge 2024-25: Developed a Non-touch, TFT-based infotainment cluster system for EVs
- Solved more than 200+ SQL Questions on HackerRank
- Certificate of Excellence Awardee 2024-25 (Punjab Engineering College, All-Rounder Category)
- Achiever Award Winner 2024-25 (Punjab Engineering College)